Rohan Kumar

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Education

University of Illinois at Urbana-Champaign **Expected Graduation Date: December 2026**
Bachelor of Science in Computer Science and Statistics (GPA: 3.86 / 4.00) *Urbana-Champaign Area, IL*
Courses: Object Oriented Programming, Data Structures, Algorithms, Database Systems, Distributed Systems, High Frequency Trading Technology, Algorithmic Market Microstructure, Statistical Modeling, Machine Learning, Machine Learning Systems

Skills

Programming Languages: Python, TypeScript, JavaScript, Java, C/C++, C#, SQL, R, HTML/CSS
Frameworks & Libraries: React, Node.js, .NET, Flask, Django, TensorFlow, Scikit-Learn, Pandas, Numpy, Prisma
Cloud & DevOps: AWS (EC2, Glue), Google Cloud Platform (GCP), Azure, Docker, Git, GitHub Actions, Spark, CI/CD
Tools & Concepts: LLMs, Multi-Agent Systems, Harnesses, Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone), Distributed Systems, RESTful APIs, Test-Driven Development, Agile, IDOR, Threat Modeling

Experience

Software Engineer Intern May 2026 – August 2026
Pinterest *New York, NY*
- Built a multi-agent AI system (TypeScript, LangGraph) that autonomously searches Pinterest's source code for security vulnerabilities, grounded in a retrieval (RAG) layer over the company's own history of confirmed vulnerabilities−surfacing a Critical-severity vulnerability that triggered a security incident and ≈12 low-medium severity bugs as of August 2026.
- Engineered a second agentic harness that dynamically validates suspected vulnerabilities by driving a Chrome browser, logging into real accounts, attempting to perform the claimed attack, and returning a Reproduced/Refuted verdict.
- Owned the platform E2E as the sole engineer building the infrastructure beneath both systems: a fleet-wide rate governor and a Postgres-backed job queue that keep hours-long agent runs alive and durable under a shared LLM token budget.
- **Technologies:** TypeScript, Python, PostgreSQL, Prisma, LangGraph, DeepAgents, RAG, LLMs, Docker, REST APIs, Git

Software Development Engineer Intern February 2026 – April 2026
Amazon *Seattle, WA*
- Designed and deployed multilingual personalization features for Amazon Music's voice recommendation system (≈23M+ daily requests), integrating user listening behavior into an ML ranking pipeline and achieving 96%+ feature coverage.
- Owned end-to-end system design and development of a language-aware candidate filtering system in Java, reducing irrelevant cross-language recommendations and improving music recommendation quality across 17M+ daily voice requests.
- Engineered 5 large-scale PySpark data pipelines (AWS Glue) to analyze 100M+ recommendation events, uncovering feature coverage gaps and critical quality issues that directly informed ranking model inputs and system design decisions.
- **Technologies:** Java, PySpark, AWS (Glue), Reinforcement Learning (RL), A/B Testing, Distributed Systems

Software Engineer Intern May 2025 – August 2025
Relativity *Chicago, IL*
- Owned the end-to-end development of internal API extensions and automated GitHub Actions CI/CD workflows, reducing direct client data access during incident resolution by 20%+ and improving resolution speed.
- Designed and engineered fault-tolerant .NET (C#) migration jobs to transition over 1TB of data from legacy SQL systems to a distributed, Azure-hosted NoSQL architecture, enhancing horizontal scalability and reducing storage costs.
- **Technologies:** C#, .NET, Azure (AKS), Kubernetes, Docker, MySQL, NoSQL, REST APIs, GitHub Actions

Projects

PAUL LLM Assistant – Illinois Business Consulting (IBC) | *Technical Lead*
- Designed, developed, and containerized an internal LLM assistant using Python and Docker to enhance workflow efficiency for 250+ active consultants by intelligently answering complex organizational queries.
- Engineered a prompt classification system to route queries to either the OpenAI API or a Pinecone vector database, enabling semantic search across past project data to generate context-aware, case-relevant responses.
- **Technologies:** Python, OpenAI API, Pinecone, Scikit-Learn, FastAPI, Docker, Large Language Models (LLMs)

Market Data Feeds Compression (PCAP Compression) – FinTech Lab @ UIUC | *Software Engineer*
- Designed and implemented a high-performance C++17 market data system for IEX TOPS/DEEP and NASDAQ ITCH PCAP feeds, using Zstandard (Zstd) with dictionary training to reduce data size by 80-85% (≈15% gain over generic Zstd).
- Engineered a streaming, memory-efficient architecture (≤ 256MB) capable of processing multi-GB captures at 40K+ packets/sec, suitable for continuous market-hours ingestion while maintaining fast decompression and system reliability.
- **Technologies:** C++17, libpcap, Zstandard (Zstd), Make, Linux, Data Infrastructure, High Frequency Trading (HFT)

Leadership

Technology Director May 2025 – December 2025
National Organization for Business and Engineering (NOBE): Illinois Chapter *Urbana-Champaign Area, IL*
- Led a team of 40+ student software engineers across 4+ technical projects for a diverse portfolio of clients.